MA1522 Linear Algebra for Computing

Lesson 14: Orthogonal Projection and the Gram-Schmidt Process

Announcement: Online Quiz 3

- **Quiz 3 covers Chapter 3**, Sections 3.1–3.8.

- It opens tomorrow, **Saturday 3 October, 0:00**, and is due **Sunday 4 October, 23:59**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson14` (e.g. `A1234567Z_lesson14.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 14**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

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Example 1

True or false? Let V be a subspace of $\mathbb{R}^n$ and $\mathbf{w} \in \mathbb{R}^n$. If

$$\mathbf{w} = \mathbf{v}_p + \mathbf{v}_n = \mathbf{u}_p + \mathbf{u}_n, \qquad \text{where } \mathbf{v}_p, \mathbf{u}_p \in V \text{ and } \mathbf{v}_n, \mathbf{u}_n \in V^\perp,$$

then

1. $\mathbf{v}_p = \mathbf{u}_p$;
2. $\mathbf{v}_n = \mathbf{u}_n$.

Both true.

$v_p + v_n = u_p + u_n \iff v_p - u_p = u_n - v_n = v \in V \cap V^\perp$.

($v_p - u_p \in V$, $u_n - v_n \in V^\perp$)

Then $v \cdot v = 0 \implies v = 0$ ($v \in V$, $v \in V^\perp$)

We conclude that $0 = v_p - u_p = v_n - u_n$.

Section 5.3 Orthogonal Projection

To see why $w - w_p$ lies in $V^\perp$.

Exercise: $w \cdot u_i = w_p \cdot u_i$.

The exercise implies that $(w - w_p) \cdot u_i = 0, \forall i$.

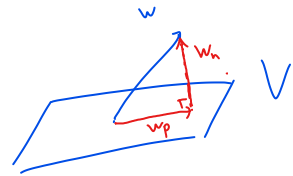
so $w_n := w - w_p \in V^\perp$.

Let V be a subspace of $\mathbb{R}^n$.

Theorem (Orthogonal projection theorem)

Every vector $\mathbf{w}$ in $\mathbb{R}^n$ can be decomposed uniquely as a sum $\mathbf{w} = \mathbf{w}_p + \mathbf{w}_n$, where $\mathbf{w}_p \in V$ and $\mathbf{w}_n \perp V$. Moreover, if $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ is an orthogonal basis for V, then

$$\mathbf{w}_p = \frac{\mathbf{w} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{w} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2}\mathbf{u}_2 + \cdots + \frac{\mathbf{w} \cdot \mathbf{u}_k}{\mathbf{u}_k \cdot \mathbf{u}_k}\mathbf{u}_k.$$



Definition

$\mathbf{w}_p$ is the **orthogonal projection** (or just **projection**) of $\mathbf{w}$ onto V.

Theorem (Best approximation theorem)

$\mathbf{w}_p$ is the vector in V closest to $\mathbf{w}$: $\|\mathbf{w} - \mathbf{w}_p\| \leq \|\mathbf{w} - \mathbf{v}\|$ for all $\mathbf{v} \in V$.

Question 1

(Hint: The formula in the previous slide start with an orthogonal basis of V.)

Let $V = \text{span}\{\mathbf{a}_1, \mathbf{a}_2\} \subseteq \mathbb{R}^3$ and $\mathbf{w} \in \mathbb{R}^3$, where

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}, \qquad \mathbf{a}_2 = \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 5 \\ -3 \\ 7 \end{pmatrix}.$$

1. Write $\mathbf{w} = \mathbf{w}_p + \mathbf{w}_n$, where $\mathbf{w}_p \in V$ and $\mathbf{w}_n \perp V$.

2. Find the smallest possible value of $\|\mathbf{w} - \mathbf{v}\|$ for $\mathbf{v} \in V$, and the $\mathbf{v}$ that attains it.

Question 1 — solution

$$S = \left\{ a_1 = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \right\} \qquad w = \begin{pmatrix} 5 \\ -3 \\ 7 \end{pmatrix}.$$

a) Step 1: Check S is orthogonal?

$$a_1 \cdot a_2 = 1 \cdot 1 + (-3) \cdot (-1) + 2(-2)$$

$$= 1 + 3 - 4 = 0$$

$\Rightarrow$ S is already orthogonal.

Step 2 Use the formula for projection.

$$w_p := \frac{w \cdot a_1}{a_1 \cdot a_1} a_1 + \frac{w \cdot a_2}{a_2 \cdot a_2} a_2$$

$$= \frac{5 + 9 + 14}{1 + 9 + 4} a_1 + \frac{5 + 3 - 14}{1 + 1 + 4} a_2.$$

$$= 2 a_1 + (-1) a_2 = \begin{pmatrix} 2 - 1 \\ -6 + 1 \\ 4 + 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ 6 \end{pmatrix}.$$

$$w_n := w - w_p = \begin{pmatrix} 5 \\ -3 \\ 7 \end{pmatrix} - \begin{pmatrix} 1 \\ -5 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}.$$

• Check $w_n \in V^\perp$

.. $w_n \cdot a_1 = 4 - 6 + 2 = 0$

.. $w_n \cdot a_2 = 4 - 2 - 2 = 0$ ✓

b) Apply thm:

$$\min_{v \in V} \{ \| w - v \| \} = \| w - w_p \| = \| w_n \| = \sqrt{4^2 + 2^2 + 1^2} = \sqrt{21}.$$

Section 5.3 Gram-Schmidt Process

Gram-Schmidt process

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ be a linearly independent set. Let

$$\mathbf{v}_1 = \mathbf{u}_1,$$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{v}_1 \cdot \mathbf{u}_2}{\|\mathbf{v}_1\|^2}\mathbf{v}_1,$$

$$\mathbf{v}_3 = \mathbf{u}_3 - \left(\frac{\mathbf{v}_1 \cdot \mathbf{u}_3}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 + \frac{\mathbf{v}_2 \cdot \mathbf{u}_3}{\|\mathbf{v}_2\|^2}\mathbf{v}_2\right),$$

$$\vdots$$

$$\mathbf{v}_k = \mathbf{u}_k - \left(\frac{\mathbf{v}_1 \cdot \mathbf{u}_k}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 + \cdots + \frac{\mathbf{v}_{k-1} \cdot \mathbf{u}_k}{\|\mathbf{v}_{k-1}\|^2}\mathbf{v}_{k-1}\right).$$

Then $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is an orthogonal set of nonzero vectors with $\text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$, and $\left\{\frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}, \ldots, \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|}\right\}$ is an orthonormal basis for it.

If S is not linearly independent, omit every $\mathbf{v}_i$ that is $\mathbf{0}$ (and leave it out of the later steps).

Question 2

Let $V = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\} \subseteq \mathbb{R}^4$, where

$$\mathbf{u}_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix}, \qquad \mathbf{u}_2 = \begin{pmatrix} 3 \\ -1 \\ 2 \\ 2 \end{pmatrix}, \qquad \mathbf{u}_3 = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 2 \end{pmatrix}, \qquad \mathbf{u}_4 = \begin{pmatrix} 1 \\ 1 \\ -4 \\ 3 \end{pmatrix}.$$

Apply the Gram-Schmidt process to $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ to obtain an orthogonal basis for V.

Question 2 — solution: $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

$u_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix}$ $u_2 = \begin{pmatrix} 3 \\ -1 \\ 2 \\ 2 \end{pmatrix}$ $u_3 = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 2 \end{pmatrix}$ $u_4 = \begin{pmatrix} 1 \\ 1 \\ -4 \\ 3 \end{pmatrix}$.

- $v_1 = u_1$, $\|v_1\|^2 = 2^2 + 1^2 + 2^2 = 9$.
- $v_2 = u_2 - \frac{u_2 \cdot v_1}{\|v_1\|^2} v_1 = \begin{pmatrix} 3 \\ -1 \\ 2 \\ 2 \end{pmatrix} - \frac{6 - 1 + 4}{9} \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix}$.

  ($\|v_2\|^2 = 1 + (-2)^2 + 2^2 = 9$. $v_1 \cdot v_2 = 2 - 2 + 0 + 0 = 0$ ✓)
- $v_3 = u_3 - \frac{u_3 \cdot v_1}{\|v_1\|^2} v_1 - \frac{u_3 \cdot v_2}{\|v_2\|^2} v_2 = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 2 \end{pmatrix} - \underbrace{\frac{2 + 3 + 4}{9}}_{1} \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix} - \underbrace{\frac{1 - 6 - 4}{9}}_{-1} \begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix}$

$$= \begin{pmatrix} 1 & -2 & +1 \\ 3 & -1 & -2 \\ -2 & +0 & +2 \\ 2 & -2 & +0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

This means $u_3 \in \text{Span}(\{v_1, v_2\}) = \text{span}(\{u_1, u_2\})$. In other words, $\{u_1, u_2, u_3\}$ is NOT linearly independent.

So we ignore u_3, and v_3 in the rest of the computation.

Question 2 — solution: $\mathbf{v}_4$

$$v_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix}, \quad u_4 = \begin{pmatrix} 1 \\ 1 \\ -4 \\ 3 \end{pmatrix}.$$

- $v_4 = u_4 - \frac{u_4 \cdot v_1}{\|v_1\|^2} v_1 - \frac{u_4 \cdot v_2}{\|v_2\|^2} v_2 - \frac{u_4 \cdot v_3}{\|v_3\|^2} v_3$ ignore this term.

$$= \begin{pmatrix} 1 \\ 1 \\ -4 \\ 3 \end{pmatrix} - \frac{2+1+6}{9} \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix} - \frac{1-2-8}{9} \begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & -2 & +1 \\ 1 & -1 & -2 \\ -4 & -0 & +2 \\ 3 & -2 & +0 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix}.$$

Answer:

$$\left\{ v_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix}, \quad v_4 = \begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix} \right\}$$ forms an orthogonal basis of $\operatorname{span}\{u_1, u_2, u_3, u_4\}$.

Alternatively, we can use ref to show that $\operatorname{span}(\{u_1, u_2, u_4\}) = \operatorname{span}(\{u_1, u_2, u_3, u_4\})$ and $\{u_1, u_2, u_4\}$ is l.I.

Then apply Gram-Schmidt to $\{u_1, u_2, u_4\}$.

Question 3

Let V and $\mathbf{u}_1, \ldots, \mathbf{u}_4$ be as in Question 2, and let $\mathbf{w} = \begin{pmatrix} 3 \\ -1 \\ -7 \\ 2 \end{pmatrix}$.

$\left\{ \begin{pmatrix} 3 \\ 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix} \right\}$

1. Using the orthogonal basis for V you found in Question 2, find the projection $\mathbf{w}_p$ of $\mathbf{w}$ onto V.

$\left\{ \begin{pmatrix} 3 \\ -1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix} \right\}$

2. The set $T = \left\{ \mathbf{u}_2, \ \mathbf{u}_3, \ \begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix} \right\}$ is another orthogonal basis for V. Use T to compute the projection of $\mathbf{w}$ onto V again, and compare with part 1.

Question 3 — solution: part 1

$$S = \left\{ v_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -2 \\ 2 \\ 0 \end{pmatrix}, \quad v_4 = \begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix} \right\}. \qquad w = \begin{pmatrix} 3 \\ -1 \\ -n \\ 2 \end{pmatrix}.$$

$\|v_1\|^2 = 9 \qquad \|v_2\|^2 = 9 \qquad \|v_4\|^2 = 9.$

$$w_p = \frac{w \cdot v_1}{\|v_1\|^2} v_1 + \frac{w \cdot v_2}{\|v_2\|^2} v_2 + \frac{w \cdot v_4}{\|v_4\|^2} v_4.$$

$$= \frac{6 - 1 + 4}{9} v_1 + \frac{3 + 2 - 14}{9} v_2 + \frac{2 + 14 + 2}{9} v_4.$$

$$= \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \\ -2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -4 \\ -4 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -6 \\ 4 \end{pmatrix}. \quad \#$$

In general, given $V = \text{span}(S)$.

To compute w_p.

Step 1. Apply Gram-Schmidt to S to get an orthogonal basis T.

Step 2 Apply the projection formula with the orthogonal basis T.

Question 3 — solution: part 2

u_2 u_3 t

$$T = \left\{ \begin{pmatrix} 3 \\ -1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ -2 \\ 1 \end{pmatrix} \right\} \qquad w = \begin{pmatrix} 3 \\ -1 \\ -7 \\ 2 \end{pmatrix}.$$

$$w_p = \frac{w \cdot u_2}{\|u_2\|^2} u_2 + \frac{w \cdot u_3}{\|u_3\|^2} u_3 + \frac{w \cdot t}{\|t\|^2} t.$$

$$= \frac{9+1-14+4}{9+1+4+4} u_2 + \frac{3-3+14+4}{1+9+4+4} u_3 + \frac{2+14+2}{4+4+1} t$$

(0; 18; 18; 18; 9)

$$= u_3 + 2t = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ -4 \\ -4 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -6 \\ 4 \end{pmatrix}$$

the same as previous part.

Next lesson

Next lesson (**Tue 6 Oct**, Sections 5.4 & 5.5), I will focus on

1. the QR factorization of a matrix with linearly independent columns, and how to compute it by the Gram-Schmidt process;

2. the definition of a least squares solution of a linear system $A\mathbf{x} = \mathbf{b}$, and how to compute it.

Please watch the video clips beforehand.